

AUTOMATION & WORKFLOW TOOLS — ASSIGNMENT SUBMISSION

# Exploring n8n: Case Studies & Template Recreation

Assignment 1 — Enterprise Case Study Analysis &  
Assignment 2 — Template Exploration and Workflow  
Recreation

|                  |                                              |
|------------------|----------------------------------------------|
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| Platform studied | <b>n8n.io</b>                                |
| Sources          | <b>n8n.io/case-studies, n8n.io/workflows</b> |

# Assignment 1 — Exploring n8n Case Studies

n8n is a source-available, fair-code workflow automation platform that lets technical and non-technical users connect apps, APIs, databases, and AI models using a visual, node-based canvas, while still allowing custom JavaScript/Python code where needed. To understand how real organizations use it in production, I reviewed the full [n8n.io/case-studies](#) library (31 published case studies spanning telecom, e-commerce, HR-tech, insurance, fintech, media, and IT operations) and then studied four in depth: **Vodafone**, **Trendyol**, **StepStone**, and **Delivery Hero**. These were chosen because together they cover the four most common enterprise patterns I saw repeated across the library: security/IT operations automation, democratizing automation across a large non-technical workforce, scaling from a grassroots pilot to an enterprise-wide platform, and a small, high-ROI single workflow with a fast turnaround.

## 1.1 Landscape Overview — What the 31 Case Studies Have in Common

Skimming across all 31 published stories, a small number of recurring themes stood out regardless of industry:

- **Grassroots adoption, then enterprise rollout.** Almost every story (Vodafone, Trendyol, StepStone, Musixmatch, Fullscript) starts with one engineer or a small team trying n8n informally, and only later does the organization move to a paid Enterprise plan for SSO, RBAC, and audit logging once usage scales past a handful of workflows.
- **The bottleneck being solved is almost always "too much manual, repetitive work relative to available engineering time."** Whether it's a CSOC triaging security alerts (Vodafone), an HR team fielding repeat questions (XIBIX), or IT resetting locked accounts (Delivery Hero), the shape of the problem is the same: high volume, low variability, and previously solvable only by adding headcount.
- **AI is now the default add-on, not an afterthought.** Later-published case studies (Trendyol's legal bot, StepStone's Confluence/Jira chatbot, Seguros Bolívar, Huel) pair n8n's orchestration with an LLM node or AI Agent to add judgment/classification/summarization on top of deterministic automation.
- **ROI is measured in hours and money, and it is large relative to the effort invested.** Several examples reported multi-day builds that then saved hundreds of hours per month or hundreds of thousands of pounds/dollars per year.

## 1.2 Deep-Dive Case Studies

Vodafone

£2.2M saved · 33 workflows

Problem / Use Case

Vodafone's Cybersecurity Operations Centre (CSOC) processes 3-5 billion security events every month. The UK's 2024 Telecom Security Act also forced longer log retention (90 days → 13 months) and wider monitoring coverage, which meant even more alerts to triage manually. The existing approach could not scale without a proportional increase in headcount.

Why / Where n8n Was Implemented

Vodafone evaluated dedicated SOAR tools (IBM Resilient, Tines) but found them too narrow. n8n was chosen because, in the words of Engineering Manager Claire Van Hinsbergh, "it provides SOAR capability and workflow capability in a low-code model, with the ability to code complex workflows... it did everything we wanted, all in one tool." It was rolled out across the CSOC, security engineering, threat intelligence, and onboarding/content teams, with implementation partner Bounteous accelerating the build-out.

How the Workflow Was Implemented

Vodafone built a library of modular, reusable workflow components (e.g., a fraud-detection module that enriches alerts with IP geolocation) stored under version control, so any team could assemble new workflows from vetted building blocks rather than starting from scratch. Workflows integrate with the CSOC ticketing system, email, and external threat-intelligence feeds, and one flow polls critical feeds every five minutes for instant alerting.

Real-World Impact

Since August 2024: 33 production workflows, 5,000+ person-days saved, £2.2M in avoided cost, and roughly £300k/month in recurring savings in 2025 — while also meeting the new regulatory retention requirements. Vodafone is now exploring agentic-AI SOC workflows as the next step.

Trendyol

700+ workflows · 1,000+ users

Problem / Use Case

Trendyol is Turkey's largest e-commerce marketplace (300,000+ sellers, 10,000+ employees, ~200 engineering teams). Every new automation idea — even something as simple as a Slack notification or a daily report — had to compete for scarce engineering time, and non-technical staff had no way to build their own solutions.

Why / Where n8n Was Implemented

Head of Engineering Hüseyin Ulaş discovered n8n on GitHub, ran a local proof-of-concept, and rolled it out to a pilot team. Within a month the community edition's lack of SSO and audit logging became a blocker for a company of Trendyol's size, so they upgraded to Enterprise. It is now self-hosted with roughly 200 isolated, team-scoped projects (one per technology team) with vault-based credential storage and daily security reviews.

How the Workflow Was Implemented

Three flagship workflows illustrate the range of use: (1) a **seller chatbot** embedded in the seller panel that routes natural-language questions through an orchestrator workflow into ~10 backend n8n workflows that run SQL/price-analysis queries; (2) a **legal AI bot** in Slack that performs retrieval-augmented generation (RAG) over internal legal documentation and cross-checks answers across multiple LLMs before the legal team validates them; and (3) a **search-relevancy agent** that periodically scores the top 40 results for each query across four storefronts and auto-files a ticket whenever relevancy drops below 80%.

Real-World Impact

In under a year: 1,000+ active users, 700+ active production workflows, and ~500,000 workflow executions per quarter. Beyond the numbers, leadership highlighted a cultural shift — non-technical employees began "thinking about their daily tasks as flow diagrams" once they saw the drag-and-drop builder.

StepStone

2-3M docs/month · 700+ workflows

Problem / Use Case

StepStone (one of Europe's largest job platforms, ~3,000 employees, ~1,000 developers globally) had used n8n's Community Edition informally since 2021, eventually processing 2-3 million job-related documents a month through hundreds of ad-hoc workflows. At that scale, shared credentials, unclear workflow ownership, weak execution monitoring, and a single master account handling GDPR-regulated data became serious liabilities. Tech Lead Luka Pilic summarized it as "effective for initial use, [but it] quickly becomes restrictive under a heavy workload."

Why / Where n8n Was Implemented

In early 2025 StepStone migrated to n8n Enterprise specifically to fix the governance gaps: SSO via Okta, an external secret store, role-based access control, and a fully separate, locked-down instance for IT workflows touching sensitive employee data.

How the Workflow Was Implemented

Beyond migrating existing workflows into governed, team-owned projects, StepStone built an internal chatbot wired to n8n's own MCP (Model Context Protocol) server plus Confluence and Jira, so employees can ask in plain language whether an automation idea is technically feasible before a workflow is even built. They also use Claude to auto-generate intentionally imperfect "starter" workflows, which nudges new users to learn the tool by finishing and fixing them rather than copy-pasting a finished flow.

Real-World Impact

Active production workflows grew from ~200 to 700+ in a year. A customer-service batch process that mapped 60-70 contracts dropped from several hours to 20 seconds. A discontinued-product-removal workflow saved the equivalent of two full working days across five full-time employees on a 4,500-row task. Multi-day manual API access requests became instant.

Delivery Hero

Built in 5 hrs · 200 hrs/mo saved

Problem / Use Case

Delivery Hero operates in 70+ countries with 53,000 employees. About 800 employees per month were getting locked out of their accounts, and every recovery required IT to manually verify and restore access across Okta and Google Workspace — averaging 35 minutes per case.

Why / Where n8n Was Implemented

The Global IT Service Delivery team (led by Dennis Zahrt, with engineer Slimani Ghaith and analyst Dorina Ababii) redesigned the process rather than just automating the old one: instead of IT approving every request, approval was shifted to the employee's own manager, and n8n was used to automate everything after that approval.

How the Workflow Was Implemented

A single n8n Enterprise workflow: it receives the manager's approval, then makes automated API calls to Okta, Jira, and Google Workspace to restore the account immediately, with no manual IT step in between. The entire workflow was designed and shipped in about five hours.

Real-World Impact

Recovery time fell from 35 to 20 minutes per incident and the team estimates roughly 200 hours saved company-wide every month. The pattern was reused for account off-boarding and software license assignment. As Zahrt put it: "We have seen drastic efficiency improvements since we started using n8n. It's incredibly powerful, but also simple to use."

### 1.3 Summary Table — Case Studies Surveyed

| Company          | Industry                    | Headline Result                                           |
|------------------|-----------------------------|-----------------------------------------------------------|
| Vodafone         | Telecom / Cybersecurity     | £2.2M saved, 5,000+ person-days, TSA compliance achieved  |
| Trendyol         | E-commerce marketplace      | 700+ workflows, 1,000+ users, ~500K executions/quarter    |
| StepStone        | HR-tech / Job platform      | 200 → 700+ workflows; hours-long tasks cut to seconds     |
| Huel             | Consumer goods (nutrition)  | Built an AI-first company culture around n8n automation   |
| Icatu Seguros    | Insurance                   | Award-winning insurance-process innovation                |
| Seguros Bolívar  | Insurance                   | 3,000 employees empowered as an "AI-first" insurer        |
| Quid             | Software / Engineering      | Aligned engineering and business to speed up delivery     |
| Delivery Hero    | Food delivery               | 200 hrs/month saved via one IT workflow, built in 5 hours |
| Fullscript       | Health-tech                 | Organization-wide AI literacy program built on n8n        |
| Deda.Tech        | IT services                 | ITSM workflows built in 30 minutes instead of 2 days      |
| Musixmatch       | Media / Music tech          | 47 days of engineering work saved in 4 months             |
| TMNZ             | FinTech                     | Productivity and innovation gains via n8n + AI            |
| iMi digital GmbH | Digital commerce agency     | Shopware imports cut from 1 day to 10 minutes             |
| K33              | Crypto / Finance            | Automated compliance and anti-money-laundering checks     |
| pxtra            | SaaS / Integrations         | Customer integrations cut from months to days             |
| Koralplay        | Gaming / Payments           | 70% of payment support tickets automated                  |
| Field Aerospace  | Aerospace / Gov contracting | Faster proposal generation for government solicitations   |
| Flow AI          | Real estate tech            | Realtor outreach cut from hours to seconds                |
| SanctifAI        | AI services                 | Human-in-the-loop review embedded into AI pipelines       |
| BeGlobal         | Consulting                  | Commercial proposals generated in under a minute          |

|                  |                             |                                                   |
|------------------|-----------------------------|---------------------------------------------------|
| XIBIX Solutions  | IT consulting               | HR question volume cut 50% with AI workflows      |
| Oversee          | Trust & safety / Support    | Automated support investigations                  |
| ITNT Media Group | Media                       | AI-driven day-to-day business operations          |
| Formula Bot      | Solo-founder SaaS           | Scaled a one-person business through automation   |
| Unbabel          | Localization / AI           | Built an advanced peak-load management system     |
| ChatHQ           | Customer communication SaaS | Shipped an all-in-one communication app           |
| TUP              | Education                   | HR and knowledge management automated with AI     |
| System           | AI operations               | Data-entry operation time reduced by 97%          |
| Bordr            | Immigration services        | Powers a \$100K online business via automation    |
| Dropsolid        | Digital agency              | Personalized marketing campaigns at scale         |
| uProc            | Data enrichment             | Replaced custom Python scripts with n8n workflows |

Source: [n8n.io/case-studies](https://n8n.io/case-studies) (accessed September 2026). Figures and quotes are as published by n8n and the featured companies.

# Assignment 2 — Exploring & Building an n8n Template

## 2.1 Browsing the Templates Library

The n8n Templates section ( [n8n.io/workflows](https://n8n.io/workflows) ) currently lists **11,800+** community- and n8n-authored workflow templates, organized into categories such as **AI, Sales, IT Ops, Marketing, Document Ops, Support**, and **Other**, and filterable by the apps they integrate with (Gmail, Google Sheets, Slack, OpenAI, Postgres, Notion, Telegram, Discord, and hundreds more). Browsing across categories, a few patterns of "shape" repeat constantly:

- **Trigger → Enrich/Classify → Act → Log** — the most common shape, seen in support, sales, and ops templates alike.
- **Scheduled batch jobs** — a Cron/Schedule trigger pulls rows from a sheet or database, transforms them, and pushes them somewhere else (reporting, syncing, cleanup).
- **AI Agent-centred workflows** — a chat trigger or webhook feeds an AI Agent node that has tools (HTTP Request, Google Sheets, Vector Store, Code) attached to it, letting the LLM decide which tool to call.

I sampled templates from each shape before choosing one, including an RSS-to-social-poster, a Postgres-to-Slack daily digest, an invoice-data-extraction pipeline, and several AI email-handling templates before settling on the one below.

## 2.2 Chosen Template

**Template:** "Auto-respond to Gmail inquiries using OpenAI, Google Sheet & AI Agent"  
**Category:** AI / Support  
**Link:** [n8n.io/workflows/4413-auto-respond-to-gmail-inquiries-using-openai-google-sheet-and-ai-agent/](https://n8n.io/workflows/4413-auto-respond-to-gmail-inquiries-using-openai-google-sheet-and-ai-agent/)

**Why I chose it:** it is small enough to fully understand end-to-end in one sitting, yet it exercises almost every core n8n concept at once — an app trigger, an LLM-based classification step, conditional branching, retrieving grounding data from a spreadsheet, an AI Agent with a system prompt, sending a reply back through the same channel, and writing an audit log. It is also directly comparable to real patterns from Assignment 1 (Trendyol's seller chatbot and StepStone's document pipelines both follow the same "classify → ground with real data → generate → log" shape), which made it a good bridge between the two assignments.

## 2.3 How the Original Template Works

**Problem it solves:** a business inbox receives a mix of genuine customer questions and non-actionable mail (newsletters, receipts, calendar invites). Manually triaging and answering the genuine questions is slow and inconsistent, but replying to everything automatically would be wrong and unprofessional. The template solves both problems: it screens out non-inquiries and only auto-answers real questions, grounded in a maintained FAQ sheet so the AI can't invent policy.

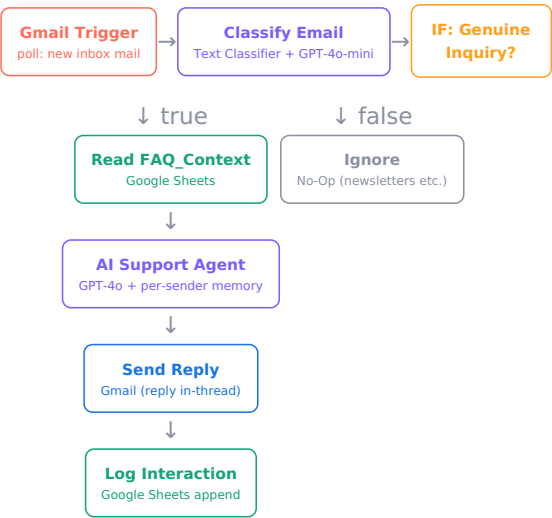
**Original node sequence (as published):**

1. **Gmail Trigger** — fires on new inbox mail.
2. **GPT-4 classification node** — labels the message "Inquiry" vs "Not Inquiry."
3. **Google Sheets lookup** — pulls relevant rows from an "FAQ\_Context" tab.
4. **AI response generator** — drafts a reply using only the retrieved FAQ context.
5. **Gmail send** — replies to the original sender.
6. **Google Sheets logging** — appends the interaction (sender, content, reply, timestamp) to an "Enquiry\_Log" tab.

Required credentials: a Gmail account (trigger + send), an OpenAI API key (classification + generation), and Google Sheets access (FAQ lookup + logging). The author ships a starter spreadsheet with the two required tabs already laid out.

## 2.4 My Recreation — Architecture

I rebuilt this template's logic as a complete, importable n8n workflow (exported as JSON, included as an appendix and as a separate attached file). I kept the same six-step shape as the original but made the underlying node choices explicit and closer to n8n's current recommended building blocks: a dedicated **Text Classifier** node (rather than a raw prompt) for the Inquiry/Not-Inquiry decision, a proper **IF** node to branch on that classification, and a LangChain **AI Agent** node (with its own chat model and a per-sender conversation memory) for the actual reply generation, so that a customer who emails twice gets a reply that remembers the earlier exchange.



## 2.5 Node-by-Node Explanation (Why Each Node Is Needed)

| # | Node           | n8n Node Type  | Purpose / Why It's There                                                                                                                                                                                                                                                  |
|---|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Gmail Trigger  | gmailTrigger   | Entry point. Polls the inbox for unread mail so the whole flow only runs when there's actually a new message — without it, nothing downstream ever fires.                                                                                                                 |
| 2 | Classify Email | textClassifier | Prevents the biggest failure mode of "auto-reply everything": accidentally answering a newsletter or shipping confirmation. Separating classification from generation also keeps the (cheaper/faster) triage step independent of the (slower/pricier) reply-writing step. |

|   |                         |                       |                                                                                                                                                                                                      |
|---|-------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | IF Genuine Inquiry?     | if                    | Turns the AI's text label into a hard branch in the workflow, so non-inquiries are cheaply short-circuited before touching the spreadsheet or the more expensive reply-generation model.             |
| 4 | Ignore (Not an Inquiry) | noOp                  | Explicit "do nothing" endpoint for the false branch — makes the workflow's intent readable at a glance instead of leaving a dangling branch.                                                         |
| 5 | Read FAQ Context        | googleSheets          | Grounds the AI in verified company facts (pricing, policies, hours) instead of letting the model improvise — directly reduces hallucination risk in a customer-facing reply.                         |
| 6 | AI Support Agent        | agent (LangChain)     | The reasoning core: combines the customer's question, the FAQ context, and a strict system prompt ("only use the supplied context; if unsure say a human will follow up") to draft the actual reply. |
| 7 | Conversation Memory     | memoryBufferWindow    | Keyed by sender email so a customer's follow-up email is answered with awareness of what was already said — the original template answers each email in isolation, this improvement adds continuity. |
| 8 | Send Reply              | gmail (reply)         | Closes the loop with the customer, replying in the same thread (not a new email) so the conversation stays intact in their inbox.                                                                    |
| 9 | Log Interaction         | googleSheets (append) | Creates an audit trail — every auto-answered question, and what the AI said, is recorded for human review, quality-checking, and spotting FAQ gaps over time.                                        |

## 2.6 Setup Steps to Run This in n8n

- Create the Google Sheet with two tabs — `FAQ_Context` (columns: Question, Answer, Category) and `Enquiry_Log` (columns: Timestamp, Sender, Subject, Question, AI Response) — and note its Sheet ID.
- In n8n, add credentials: Gmail OAuth2, OpenAI API key, Google Sheets OAuth2.
- Import `gmail-ai-support-responder.json` (attached) via *Workflows → Import from File*, or build the nine nodes above manually in the order shown in section 2.4.
- Open each Google Sheets node and each Gmail node and pick your credential + the real Sheet ID from the dropdown (the JSON ships with placeholder IDs).
- Write the system prompt on the AI Support Agent node (a starting version is included in the JSON).
- Send a test email to the connected inbox, then activate the workflow once the test run produces a correct reply and log row.

## 2.7 What I Learned

Two things stood out. First, the value of the classification-before-generation split: it's tempting to just prompt one big model to "reply if appropriate," but splitting triage from generation makes the workflow cheaper, easier to debug (you can see exactly which step misfired), and safer, since the expensive/creative step never runs on mail that shouldn't get a reply at all. Second, grounding the AI Agent in a live Google Sheet rather than a hard-coded prompt means non-technical staff can improve the bot's answers just by editing a spreadsheet — the same "democratization" pattern I saw repeatedly in the Assignment 1 case studies (StepStone's non-engineers building their own automations, Trendyol's business teams thinking in flow diagrams). It's a small template, but it's a genuine example of the trigger → classify → ground → generate → act → log pattern that shows up, at much larger scale, across almost every case study reviewed in Assignment 1.

## 2.8 Appendix — Exported Workflow JSON

Full n8n-importable workflow (also attached separately as `gmail-ai-support-responder.json`):

```
{
  "name": "Gmail AI Inquiry Classifier, Auto-Responder & Logger",
  "nodes": [
    {
      "parameters": {
        "pollTimes": {
          "item": [
            { "mode": "everyMinute" }
          ]
        },
        "simple": false,
        "filters": {
          "labelIds": ["INBOX"],
          "readStatus": "unread"
        },
        "options": {}
      },
      "id": "1a2b3c4d-0001-4e5f-8a6b-000000000001",
      "name": "Gmail Trigger",
      "type": "n8n-nodes-base.gmailTrigger",
      "typeVersion": 1.2,
      "position": [-1120, 300],
      "credentials": {
        "gmailOAuth2": { "id": "GMAIL_CRED_ID", "name": "Gmail account" }
      }
    },
    {
      "parameters": {
        "model": { "value": "gpt-4o-mini" },
        "options": { "temperature": 0 }
      },
      "id": "1a2b3c4d-0002-4e5f-8a6b-000000000002",
      "name": "OpenAI Chat Model (Classifier)",
      "type": "@n8n/n8n-nodes-langchain.lmChatOpenAi",
      "typeVersion": 1.2,
      "position": [-880, 520],
      "credentials": {
        "openAiApi": { "id": "OPENAI_CRED_ID", "name": "OpenAI account" }
      }
    },
    {
      "parameters": {
        "inputText": "=Subject: {{{json.subject}}}\nFrom: {{{json.from}}}\nBody:\n{{{json.text}}}",
        "categories": {
          "categories": [
            {
              "category": "Inquiry",
              "description": "A genuine customer question, support request, or business inquiry that expects a reply."
            },
            {
              "category": "Not Inquiry",
              "description": "Newsletters, marketing emails, receipts, automated notifications, calendar invites, or spam that do not need a personal reply."
            }
          ]
        },
        "options": { "fallback": "Not Inquiry" }
      },
      "id": "1a2b3c4d-0003-4e5f-8a6b-000000000003",
```

```

"name": "Classify Email (Text Classifier)",
"type": "@n8n/n8n-nodes-langchain.textClassifier",
"typeVersion": 1,
"position": [-880, 300]
},
{
  "parameters": {
    "conditions": {
      "options": { "caseSensitive": true, "leftValue": "", "typeValidation": "strict" },
      "conditions": [
        {
          "leftValue": "={{$.category}}",
          "rightValue": "Inquiry",
          "operator": { "type": "string", "operation": "equals" }
        }
      ],
      "combinator": "and"
    },
    "options": {}
  },
  "id": "1a2b3c4d-0004-4e5f-8a6b-000000000004",
  "name": "IF Genuine Inquiry?",
  "type": "n8n-nodes-base.if",
  "typeVersion": 2,
  "position": [-640, 300]
},
{
  "parameters": {
    "operation": "noOp"
  },
  "id": "1a2b3c4d-0005-4e5f-8a6b-000000000005",
  "name": "Ignore (Not an Inquiry)",
  "type": "n8n-nodes-base.noOp",
  "typeVersion": 1,
  "position": [-420, 480]
},
{
  "parameters": {
    "documentId": { "value": "FAQ_SHEET_ID" },
    "sheetName": { "value": "FAQ_Context" },
    "filtersUI": {},
    "options": {}
  },
  "id": "1a2b3c4d-0006-4e5f-8a6b-000000000006",
  "name": "Read FAQ Context (Google Sheets)",
  "type": "n8n-nodes-base.googleSheets",
  "typeVersion": 4.5,
  "position": [-420, 220],
  "credentials": {
    "googleSheetsOAuth2Api": { "id": "GSHEETS_CRED_ID", "name": "Google Sheets account" }
  }
},
{
  "parameters": {
    "model": { "value": "gpt-4o" },
    "options": { "temperature": 0.3 }
  },
  "id": "1a2b3c4d-0007-4e5f-8a6b-000000000007",
  "name": "OpenAI Chat Model (Responder)",
  "type": "@n8n/n8n-nodes-langchain.lmChatOpenAi",
  "typeVersion": 1.2,
  "position": [-180, 460],
  "credentials": {
    "openAiApi": { "id": "OPENAI_CRED_ID", "name": "OpenAI account" }
  }
},
{
  "parameters": {
    "sessionIdType": "customKey",
    "sessionKey": "={{$('Gmail Trigger').item.json.from}}",
    "contextWindowLength": 5
  },
  "id": "1a2b3c4d-0008-4e5f-8a6b-000000000008",
  "name": "Conversation Memory (per sender)",
  "type": "@n8n/n8n-nodes-langchain.memoryBufferWindow",
  "typeVersion": 1.3,
  "position": [-40, 460]
},
{
  "parameters": {
    "promptType": "define",
    "text": "=A customer emailed us. Using ONLY the verified FAQ context below, write a helpful, professional reply. If the FAQ context does not cover the question, politely say a human will follow up and do not invent facts.\n\nCustomer email:\nSubject: {{$('Gmail Trigger').item.json.subject}}\n{{$('Gmail Trigger').item.json.text}}\n\nVerified FAQ context:\n{{$$.toJsonString()}}",
    "options": {
      "systemMessage": "You are a professional, friendly customer-support assistant for the company. Keep replies concise, accurate, and grounded strictly in the supplied FAQ context. Never make up policies, prices, or dates."
    }
  },
  "id": "1a2b3c4d-0009-4e5f-8a6b-000000000009",
  "name": "AI Support Agent",
  "type": "@n8n/n8n-nodes-langchain.agent",
  "typeVersion": 1.9,
  "position": [-180, 220]
},
{
  "parameters": {
    "operation": "reply",
    "messageId": "={{$('Gmail Trigger').item.json.id}}",
    "emailType": "text",
    "message": "={{$.output}}",
    "options": {}
  },
  "id": "1a2b3c4d-000a-4e5f-8a6b-00000000000a",
  "name": "Send Reply (Gmail)",
  "type": "n8n-nodes-base.gmail",
  "typeVersion": 2.1,
  "position": [80, 220],
  "credentials": {
    "gmailOAuth2": { "id": "GMAIL_CRED_ID", "name": "Gmail account" }
  }
},
{
  "parameters": {
    "operation": "append",
    "documentId": { "value": "FAQ_SHEET_ID" },
    "sheetName": { "value": "Enquiry_Log" },
    "columns": {

```

```

    "mappingMode": "defineBelow",
    "value": {
      "Timestamp": "={{$now.toISO()}}",
      "Sender": "={{$('Gmail Trigger').item.json.from}}",
      "Subject": "={{$('Gmail Trigger').item.json.subject}}",
      "Question": "={{$('Gmail Trigger').item.json.text}}",
      "AI Response": "={{$('AI Support Agent').item.json.output}}"
    }
  },
  "options": {}
},
{
  "id": "1a2b3c4d-000b-4e5f-8a6b-00000000000b",
  "name": "Log Interaction (Google Sheets)",
  "type": "n8n-nodes-base.googleSheets",
  "typeVersion": 4.5,
  "position": [340, 220],
  "credentials": {
    "googleSheetsOAuth2Api": { "id": "GSHEETS_CRED_ID", "name": "Google Sheets account" }
  }
},
],
"connections": {
  "Gmail Trigger": {
    "main": [[{ "node": "Classify Email (Text Classifier)", "type": "main", "index": 0 }]]
  },
  "OpenAI Chat Model (Classifier)": {
    "ai_languageModel": [[{ "node": "Classify Email (Text Classifier)", "type": "ai_languageModel", "index": 0 }]]
  },
  "Classify Email (Text Classifier)": {
    "main": [[{ "node": "IF Genuine Inquiry?", "type": "main", "index": 0 }]]
  },
  "IF Genuine Inquiry?": {
    "main": [
      [{ "node": "Read FAQ Context (Google Sheets)", "type": "main", "index": 0 }],
      [{ "node": "Ignore (Not an Inquiry)", "type": "main", "index": 0 }]]
  },
  "Read FAQ Context (Google Sheets)": {
    "main": [[{ "node": "AI Support Agent", "type": "main", "index": 0 }]]
  },
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  "Conversation Memory (per sender)": {
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  "Send Reply (Gmail)": {
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